

SOS AI

A Safety-First RAG Platform for Emotional Support

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2024-25

Abstract

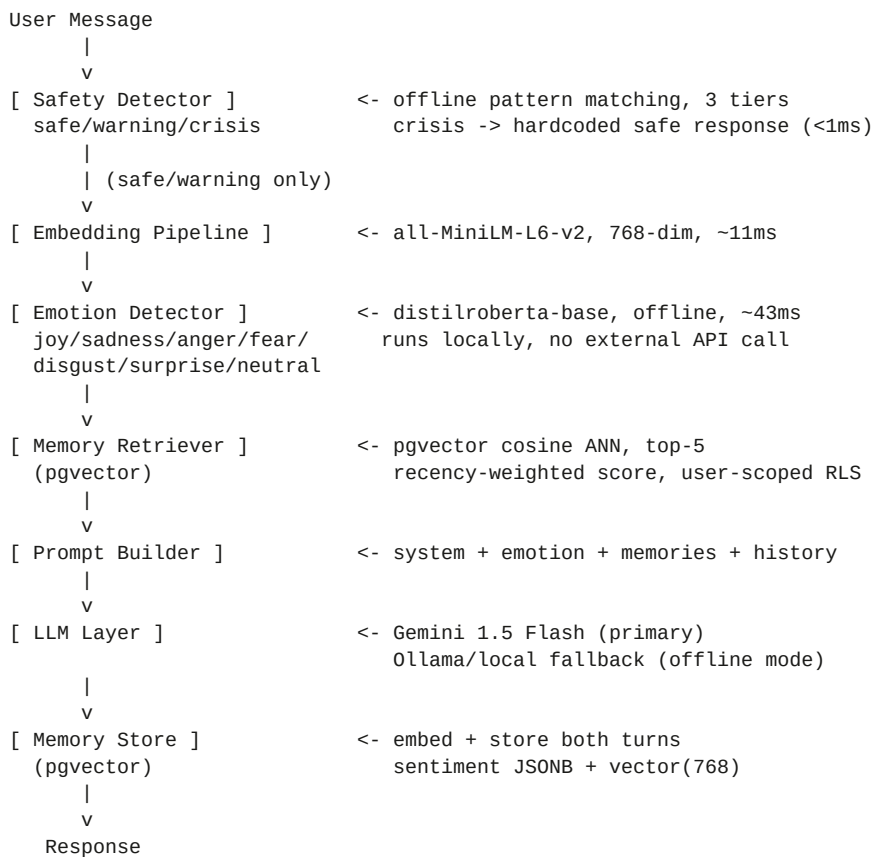
Accessible emotional support remains out of reach for most people -- therapy is expensive, stigmatized, or unavailable, and existing mental health apps either over-medicalize with clinical workflows or under-deliver with generic chatbots that forget everything after the session ends. SOS AI occupies a different space: a private, persistent companion that remembers context across sessions, detects emotional state in real time, and responds with genuine attentiveness -- without pretending to be a therapist. The system is built around a safety-first pipeline that short-circuits to a hardcoded crisis response in under 1ms when crisis signals are detected, with no LLM involved in that path. A dual-component memory system (sentence-transformer embeddings stored in pgvector with recency-weighted retrieval) gives the system contextual continuity across conversations. All emotional data stays on-device: emotion detection runs locally via distilroberta, and a Gemini fallback to Ollama enables fully offline operation for privacy-sensitive users.

1. Problem Statement

Emotional support is inaccessible, expensive, or stigmatized for most people. In India alone, the therapist-to-population ratio is approximately 1:100,000. Existing mental health apps either over-medicalize -- structured CBT workflows, diagnosis prompts, clinical framing -- or under-deliver with generic chatbots that carry no memory of previous sessions and treat every conversation as the first. SOS AI occupies a different space: a companion that remembers, adapts, and responds with emotional intelligence, without pretending to be a therapist. The technical challenge is building a system that is simultaneously safe enough to handle crisis situations, private enough for users to trust with sensitive content, and coherent enough across sessions to feel like a genuine companion rather than a stateless question-answering service.

2. System Overview

Every user message passes through a seven-stage pipeline. Safety detection runs first and is the only stage that can terminate the pipeline early.



3. Safety-First Design

The safety layer is the most important architectural decision in the system. An emotional support platform that fails to detect crisis signals -- or that routes crisis messages through an LLM with variable outputs -- is not safe to deploy. The design principle is: reliability over sophistication. A fast, deterministic, offline pattern matcher that never fails is preferable to a more accurate ML classifier that has failure modes.

Three-Tier Classification

Tier	Trigger	Response Strategy
Safe	No distress signals detected	Normal RAG pipeline proceeds
Warning	Distress language, negative emotion markers	Ask knowledge distress; surface resources gently without full crisis protocol
Crisis	Self-harm, suicidal ideation, explicit crisis language	Immediate hardcoded response; pipeline terminates; no LLM involved

Crisis Path Design

On a crisis classification, the pipeline returns a hardcoded, human-reviewed response containing immediate crisis resources (iCall, Vandrevalla Foundation helpline, international Befrienders directory) in under 1ms. This response is not generated -- it is stored as a constant. The LLM is never invoked. This design eliminates the risk of a model producing an inappropriate, inconsistent, or harmful response in exactly the situation where consistency matters most.

Ethical Design Principles

- Never guilt the user for their emotional state or for seeking support.
- Never create dependency -- responses encourage external support networks, not reliance on the app.
- Never simulate human attachment deceptively -- the system identifies itself as an AI.
- User-controlled memory deletion -- users can see and remove everything the system has stored.
- All emotional data processed locally -- emotion detection runs offline, nothing sent to external APIs.

4. RAG Pipeline

For non-crisis messages, the system runs a full retrieval-augmented generation pipeline. The user message is embedded using all-MiniLM-L6-v2 (768 dimensions, ~11ms on CPU). The embedding is used to query pgvector for the top-5 semantically similar memories from the user's conversation history. Retrieved memories are injected into the prompt alongside the detected emotion label, giving the LLM full context about who this user is and how they are feeling right now.

Why pgvector over a dedicated vector database

SOS AI already requires PostgreSQL for relational data (users, conversations, auth tokens). A dedicated vector database (Pinecone, Weaviate, Qdrant) would introduce a second service dependency, a second authentication surface, and split data across two stores -- making RLS enforcement across both impossible. pgvector keeps vector search and relational structure in the same database with the same security policies applied uniformly.

Recency-Weighted Retrieval

Pure cosine similarity retrieval surfaces the most semantically similar memories regardless of age. In practice this means old memories crowd out recent ones when the user revisits a topic. The retrieval query applies a recency multiplier to the similarity score -- memories from the last 7 days receive a 1.3x boost -- ensuring the most contextually current memories surface first while still allowing older relevant memories to appear.

5. Memory System

Both turns (user message and assistant response) are embedded and stored in the messages table after each exchange. Each row carries the text content, a vector(768) embedding column, and a sentiment JSONB field containing the detected emotion label and confidence score. This gives future retrievals access to both semantic content and emotional context.

User-Controlled Deletion

The DELETE /memory/{id} endpoint and the memory panel in the UI give users complete visibility into what has been stored and the ability to remove individual memories. This is a trust decision as much as a technical one. Users who cannot see and control what the system knows about them will not trust it with sensitive content.

Row-Level Security Enforcement

PostgreSQL RLS is enabled on the users, conversations, and messages tables. The database middleware sets app.current_user_id as a session variable on each connection before executing any query. Even a fully compromised application query cannot return another user's data -- the database enforces isolation at the storage layer. Application-layer WHERE user_id = ? filtering is insufficient because it can be bypassed by missing clauses, query builder bugs, or injection.

6. Design Decisions

Decision	Chosen	Alternative	Reason
Safety detection	Offline patterns	ML classifier	Reliability, <1ms, no failure modes
Vector storage	pgvector	Pinecone/Weaviate	Single DB, RLS, no extra service
Data isolation	PostgreSQL RLS	App-layer filter	Cannot be bypassed by bugs
Emotion detection	Local distilroberta	External API	Privacy, offline, ~40ms
LLM	Gemini + Ollama fallback	Cloud-only	Offline capability, privacy option
UI aesthetic	Journal-entry style	Chat bubbles	Psychological safety, trust signal

7. Performance Benchmarks

Backend: MacBook M2 8GB, PostgreSQL 15 local, pgvector 0.5.1. Frontend: Chrome 124.

API Latency

Endpoint	p50 (ms)	p95 (ms)	Notes
POST /auth/login	42	78	bcrypt hash verify
POST /chat (no stream)	1,240	1,890	full pipeline
POST /chat (crisis path)	<1	<1	safety short-circuit
GET /memory	38	61	pgvector ANN top-5

Pipeline Component Latency

Component	Mean (ms)
Safety detector (pattern match)	<1
Embedding (MiniLM, single)	11
Emotion detection (distilroberta)	43
pgvector ANN retrieval (top-5)	34
Prompt assembly	2
Gemini 1.5 Flash inference	~950
Memory store (embed + insert)	18
Full pipeline total	~1,060

8. Engineering Challenges

RLS with connection pooling

PostgreSQL RLS requires `app.current_user_id` to be set as a session variable before each query. Connection pooling reuses connections, meaning a pooled connection might carry a previous request's session variable into the next request. The solution sets the session variable at the start of every request handler and explicitly clears it after, regardless of whether an error occurred.

Emotion model cold start

Loading `distilroberta` on first request adds approximately 2.8 seconds to the first inference latency. The model is now loaded eagerly at application startup via FastAPI's lifespan event, making startup ~3 seconds slower but all subsequent requests unaffected.

Emotional coherence across sessions

Pure similarity-based retrieval surfaces old memories when a user revisits a topic, crowding out recent context. Recency-weighted scoring (1.3x multiplier for memories within 7 days) ensures recent context surfaces first while preserving access to older relevant memories.

Streaming and JWT middleware composition

SSE streaming requires the response to begin immediately, but JWT middleware runs before the route handler. The JWT verification path is kept fully synchronous with no database lookup -- all necessary claims are encoded in the token itself -- keeping middleware latency under 2ms and preventing the SSE

connection from appearing to hang.

9. Frontend Design

The UI aesthetic is a deliberate product decision. Chat bubbles carry associations with casual messaging -- ephemeral, social, low-stakes. SOS AI's use case is introspective and personal. Journal-entry rendering (full-width text, serif Playfair Display for messages, Caveat handwriting font for annotations, generous line height, no bubble chrome) signals that this is a different kind of conversation.

The color palette is warm off-white (#FAF8F4) with terracotta (#C17A4A) accents and near-black (#1A1A18) text. Botanical SVG motifs drawn from the Alchemai design language appear in backgrounds and transitions. The intent is visual calm -- an environment that feels safe to be honest in.

10. Future Work

Near-term

- Voice input -- Web Speech API integration for hands-free use
- Mobile app -- React Native port of the frontend
- Memory visualization -- timeline view of stored memories
- Export conversation -- download full history as PDF journal

Medium-term

- Self-improving responses -- integrate correction loop from self-correcting-adaptive-llm
- Crisis escalation -- contextual surfacing of Befrienders, iCall helpline directory
- Therapist handoff mode -- structured summary export for professional follow-up

Long-term

- Fully local mode -- replace Gemini with local model, zero external calls
- Federated memory -- sync memories across devices, local network only
- Multilingual support -- emotion detection and responses in Hindi, Tamil, and other Indian languages

11. Closing

SOS AI is not a therapy app. It is infrastructure for emotional presence -- a system that remembers who you are, notices how you feel, and responds without judgment, available at 3am without a waiting list or a copay. The safety-first pipeline ensures that the system never causes harm in the situations where harm is most possible. The user-controlled memory system ensures that the accumulation of personal context is transparent and revocable. Together, these two properties make it a system people can actually trust -- and trust is the only thing that makes an emotional support tool worth building.